CMUC Online Seminar on Algebra, Logic and Topology

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A general approach to pointfree T_0 spaces

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Abstract

The classical dual adjunction between frames and spaces, given by the functors O : Top → Frm and pt : Frm → Top, restricts to a dual equivalence between sober spaces and spatial frames. Another pointfree approach to the study of spaces is the so-called T_D duality, a similar dual adjunction where the fixpoints are the T_D spaces. The two dualities are incompatible in that the two spectrum functors are different. Recently, a pointfree approach to spaces whose fixpoints are all T_0 spaces was introduced. The approach is inspired by Raney duality. In this setting, we study the category of Raney extensions and show that we have a dual adjunction extending (O, pt) and capturing all T_0 spaces. By suitably restricting on objects we recover both the classical and the T_D duality. Other pointfree notions of T_0 spaces are given by strictly zero-dimensional biframes and McKinsey-Tarski algebras. Aiming to understand the relationship between these three different settings, we introduce and study an abstract notion of pointfree T_0 space. We study the situation where we have a suitable forgetful functor U : C → Frm and a dual adjunction O_C : Top → C and pt_C : C → Top, whose fixpoints are the T_0 spaces and which is, in a sense that will be made precise, compatible with the adjunction (O, pt) between frames and spaces. By studying the fibers of U , we are able to study various aspects of T_0 spaces abstractly. In all pointfree notions of T_0 spaces mentioned above, the pointfree sober spaces are the top elements of the fibers, and the T_D ones are the bottom elements. This talk is based on ongoing joint work with Rui Prezado.